

UNIVERSITI KEBANGSAAN MALAYSIA
The National University of Malaysia

CMOS Technology: Where We Are and How We Got Here

KKKC6353 Nanoelectronic Devices — Lecture 1, Document 1 of 3

Dr Muhammad Faiz Bukhori
Universiti Kebangsaan Malaysia

Revised October 2026

Table of Contents

How this lecture is organised 3
The argument in one paragraph 3
1 From one transistor to the integrated circuit 3
1.1 The invention, and a prior claim 3
1.2 From Bell Labs to Silicon Valley 4
1.3 The integrated circuit 4
2 Moore's law: what it actually said 5
2.1 The 1965 paper 5
2.2 Three things students routinely get wrong 6
2.3 The trend through 2026 7
3 Dennard scaling: the engine underneath 8
4 The end of Dennard scaling 8
4.1 The mechanism of failure 8
4.2 What it looked like from outside 9
5 What the industry did instead 10
5.1 Strained silicon (90 nm, 2003) 11
5.2 High-k dielectric and metal gate (45 nm, 2007) 11
5.3 FinFET (22 nm, 2011) 13
5.4 Gate-all-around nanosheet (2 nm class, 2025) 13
5.5 Backside power delivery (2025 onward) 14
5.6 CFET (research; projected around 2032) 14
6 Lithography: the enabling constraint 15
7 The node-name problem 15
8 The industry in 2026 16
9 Where this leaves us 17
Sources and further reading 18

How this lecture is organised

Lecture 1 is delivered as three documents, which should be read in order:

	Document	What it answers
1	*CMOS Technology: Where We Are and How We Got Here* (this document)	What the industry built, why classical scaling stopped working, and what replaced it
2	*Device Physics for Nanoscale CMOS: A Working Revision*	The physics you need, stated once, in one notation — ending with the point where that physics breaks down
3	*Short-Channel Effects in Nanoscale CMOS*	Why small transistors misbehave, and what is done about it

This document carries no derivations. It is the argument that motivates the other two, and the rest of the module after them. You are expected to be familiar with semiconductor physics already; Document 2 is a revision, not an introduction.

The argument in one paragraph

For roughly thirty years, making transistors smaller made them simultaneously denser, faster and more power-efficient, at no cost. That was not a law of nature — it was a specific engineering recipe called Dennard scaling, and it depended on being able to reduce the supply voltage in step with the dimensions. Around 2003 that recipe failed, for a reason that is entirely a matter of device physics: the threshold voltage could no longer be scaled down without the transistor leaking current when it was supposed to be off. Everything the semiconductor industry has done since — strained silicon, high-k dielectrics, FinFETs, gate-all-around nanosheets, backside power delivery, and the CFET structures now in research — is an attempt to keep scaling density in the absence of that recipe. **Understanding why the recipe failed is the content of Documents 2 and 3. Understanding what was built in response is the content of this one.**

1 From one transistor to the integrated circuit

1.1 The invention, and a prior claim

The point-contact transistor was demonstrated at Bell Telephone Laboratories in December 1947 by **John Bardeen** and **Walter Brattain**, working in a group led by **William Shockley**. Shockley, who had not been present at the first demonstration, went on to devise the far more manufacturable *junction* transistor shortly afterwards. The three shared the 1956 Nobel Prize in Physics.



Figure 1.1. The point-contact transistor of December 1947: two gold contacts, held by a plastic wedge, pressed onto a slab of germanium. Original photograph: Bell Laboratories; reproduced from W. A. Doolittle, Georgia Tech lecture slides (2011).

The field-effect principle itself was patented much earlier — by the Austro-Hungarian physicist **Julius Edgar Lilienfeld**, who filed in Canada in October 1925 and in the United States in 1926 and 1928. Lilienfeld never built a working device; the materials science of the 1920s could not deliver a semiconductor surface clean enough. The gap between the idea and the artefact is about forty years of materials processing, and that gap is a recurring theme in this module: in nanoelectronics, the concept is almost never the bottleneck.

> **A note on the date.** The date usually quoted, 23 December 1947, is when the device was demonstrated to Bell Labs management. It first worked on **16 December 1947**.

1.2 From Bell Labs to Silicon Valley

Shockley's relationship with Bell Labs management deteriorated through the early 1950s and came to a head with his 1953 sabbatical, during which he returned to Caltech as a visiting professor. In **1955** he founded the **Shockley Semiconductor Laboratory**, funded by Beckman Instruments, which opened for business in **Mountain View, California, in 1956**. He recruited young graduates rather than established colleagues.

In **1957**, eight of them — including **Gordon Moore** and **Robert Noyce** — resigned and founded **Fairchild Semiconductor**. The precipitating disagreement was partly Shockley's management and partly his wish to move away from silicon-based work. In **1968** Moore and Noyce left Fairchild to found **Intel**.

That lineage matters for one reason only: it explains why an observation about manufacturing cost, made by a research director at Fairchild in 1965, became the organising principle of an entire industry.

1.3 The integrated circuit

Jack Kilby at Texas Instruments demonstrated the first integrated circuit in **1958**, building several components on a single piece of germanium rather than wiring discrete parts together. He received the Nobel Prize in Physics in 2000. **Robert Noyce**, independently and some months later, arrived at a silicon-based version using the **planar process** — which is the one that proved manufacturable and is the direct ancestor of everything built since. Noyce was not a co-recipient of the 2000 prize; he had died in 1990, and the prize is not awarded posthumously.

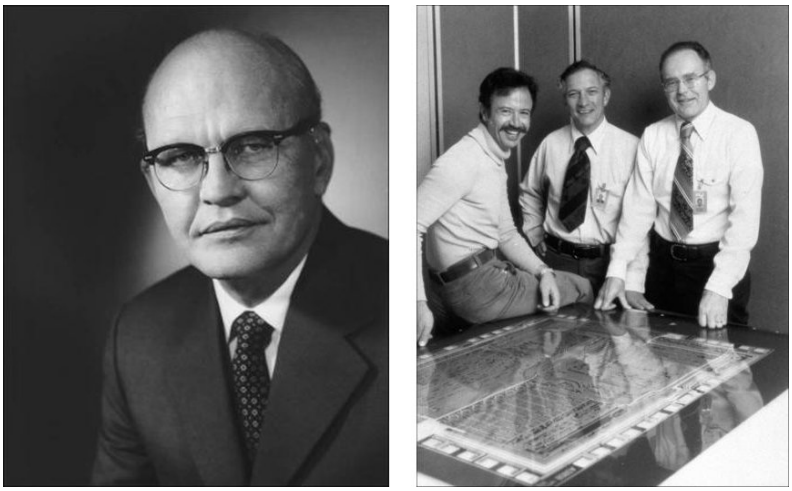


Figure 1.2. *Left: Jack Kilby of Texas Instruments, who demonstrated the first integrated circuit in 1958. Right: Andy Grove, Robert Noyce and Gordon Moore of Intel, with the layout of the Intel 8080 microprocessor. Photographs: Texas Instruments and Intel; reproduced from S. Levkov, NJIT ECE 271 lecture slides, Ch. 2.*

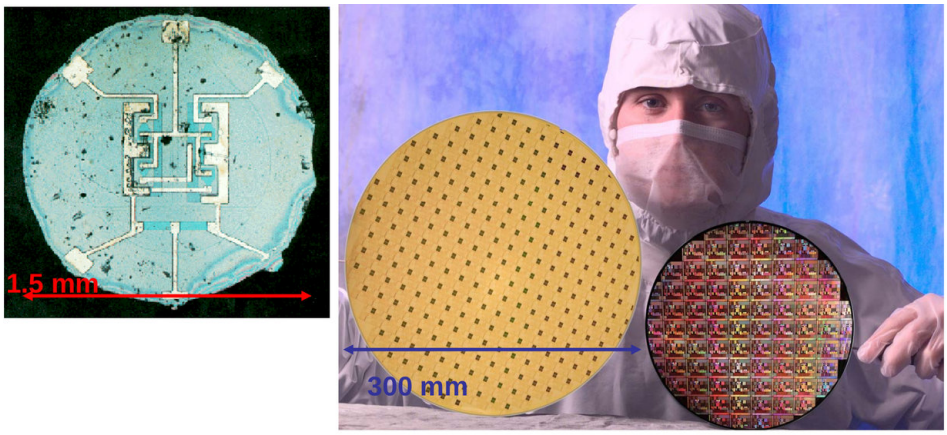


Figure 1.3. *Fifty years of the planar process. Left: Fairchild's first planar integrated circuit (1961), about 1.5 mm across. Right: 200 mm and 300 mm silicon wafers. Sources: Smithsonian Chip Collection (left), IBM (right); reproduced from W. A. Doolittle, Georgia Tech lecture slides (2011).*

2 Moore's law: what it actually said

2.1 The 1965 paper

Moore's law is named for a four-page article Gordon Moore wrote for the trade magazine *Electronics* in April 1965 (vol. 38, no. 8, pp. 114–117), while he was director of research at Fairchild. He observed that the number of components per integrated circuit had doubled roughly every year since 1958, and predicted the trend would hold for at least another decade. In **1975** he revised the rate to a **doubling every two years**.

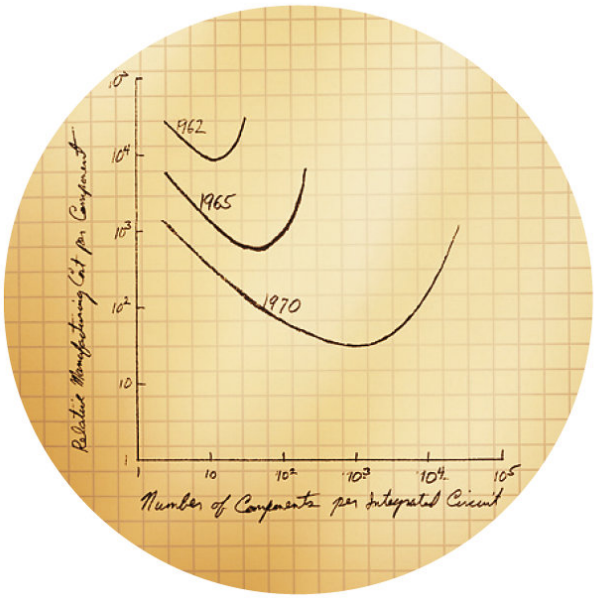


Figure 1.4. *Moore's original 1965 figure. The vertical axis is relative manufacturing cost per component; the horizontal axis is the number of components on one chip. Each curve has a minimum — the cost-optimal complexity for that year's process — and the minimum moves right and down over time. The famous doubling is a statement about where this minimum sits, not about physics. Figure from G. E. Moore, Electronics, 1965; image courtesy of Intel, as reproduced in earlier course material.*

2.2 Three things students routinely get wrong

It is an economic observation, not a physical law. Look again at the figure above: the quantity being minimised is *cost per component*. Moore's claim was that the cost-optimal number of components per chip doubles on a regular cadence. Nothing in physics requires this. It held for sixty years because an entire industry organised its capital investment around making it hold.

The exponent is not 18 months. Moore said one year in 1965 and two years in 1975. "Every 18 months" is a later paraphrase, commonly attributed to Intel executive David House, who combined transistor count with per-transistor speed gains to predict performance doubling. If you are going to quote a number, quote *two years*, and say what is doubling.

Transistor count is not performance. The two tracked each other until roughly 2004 and then separated completely. Section 4 is about that separation.

2.3 The trend through 2026

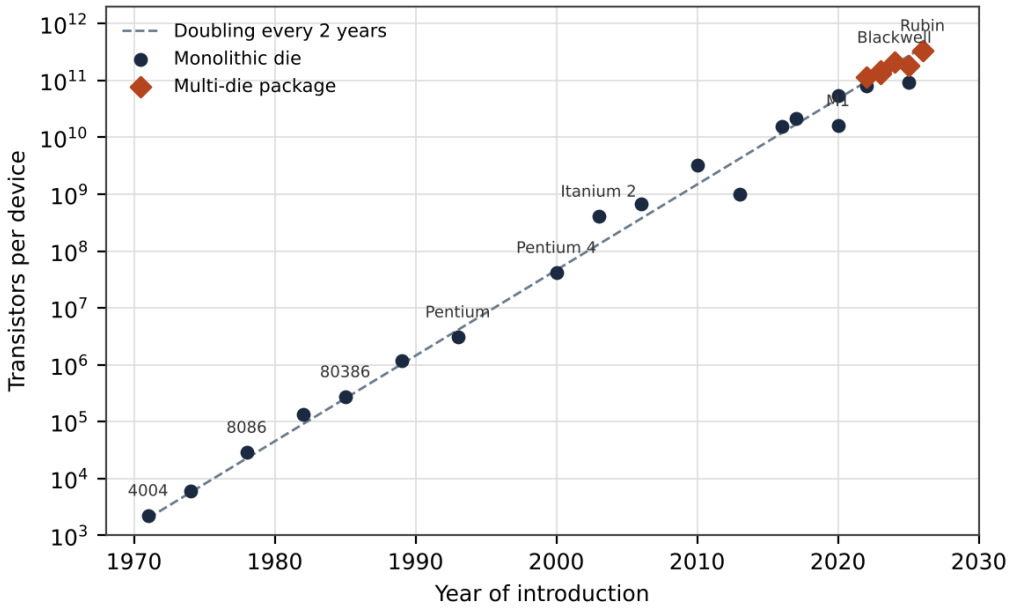


Figure 1.5. *Transistor count of landmark processors and accelerators, 1971–2026, against a two-year doubling reference. Note the distinction between monolithic dies and multi-die packages: several of the headline figures from 2022 onward are packages containing two or more separate dies, not single pieces of silicon. Data: Wikipedia, Transistor count (accessed October 2026).*

Two observations follow from this figure.

First, the trend genuinely has continued, and anyone who told you Moore's law was dead in 2016 was wrong. Second — and this is the more interesting point — the way it has continued changed. The highest-count devices on this chart are multi-die packages: NVIDIA's two-die Blackwell (208 billion) and Rubin (336 billion), and Apple's Ultra parts, are assemblies of separate dies joined by advanced packaging. Counting those as "one chip" is a defensible bookkeeping choice, but it is a different claim from the one Moore made about components per *integrated circuit*. Ask yourself whether the law survives if you insist on monolithic dies only.

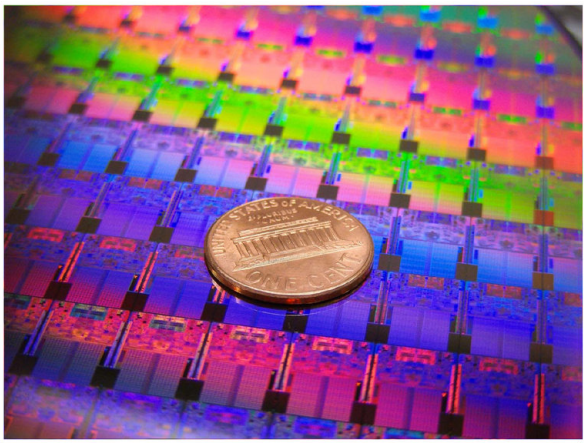


Figure 1.6. *A 300 mm wafer of Intel 45 nm-generation processors (2008), with a US one-cent coin for scale. Each rectangle is a single die carrying of the order of* 10^{8} *to* 10^{9} *transistors. Image: Intel; reproduced from W. A. Doolittle, Georgia Tech lecture slides (2011).*

3 Dennard scaling: the engine underneath

Moore's law describes *what* happened. **Dennard scaling** explains *why it was worth doing*, and its failure is the single most important fact in this module.

In 1974, Robert Dennard and colleagues at IBM published a set of rules for scaling a MOSFET. Take every dimension — channel length L, channel width W, oxide thickness t_{ox}, junction depth — and divide it by a factor $\kappa > 1$. Divide the supply voltage V_{DD} and the threshold voltage V_T by the same κ. Multiply the substrate doping N_A by κ. The consequences are remarkable:

Quantity	Scales as	Effect for $\kappa = 1.4$
Device area	$1/\kappa^2$	halved
Gate capacitance $C = \varepsilon_{ox} WL/t_{ox}$	$1/\kappa$	$\times$ 0.71
Drive current I_D	$1/\kappa$	$\times$ 0.71
Gate delay $\tau \sim CV/I$	$1/\kappa$	$\times$ 0.71 (faster)
Power per device $P = IV$	$1/\kappa^2$	halved
Power density P/area	**constant**	**unchanged**
Electric field in the device	constant	unchanged

Read the last two rows carefully, because they are the whole point. You get twice as many transistors, each one 40% faster, and **the chip does not get any hotter**. Density, speed and efficiency all improve together, and nothing has to be invented to make it happen — you simply print the same design smaller. This is why the industry could organise itself around a two-year cadence for three decades. Every node was a free lunch.

The scheme is called **constant-field scaling** because the internal electric field stays fixed: both the voltages and the distances shrink by κ.

> **How the industry actually scaled.** Dennard's recipe was not followed literally. Through the 1970s and 1980s the supply voltage was held at 5 V for compatibility with existing logic families — *constant-voltage* scaling — so the internal field rose by κ with every generation. The resulting hot-carrier damage is why the lightly doped drain (LDD) structure was introduced around 1980. Supply voltages began to fall only with the 0.5–0.35 μm generations in the 1990s (5 → 3.3 → 2.5 V → ⋯), bringing practice close to constant-field scaling until V_{DD} stalled near 1 V in the mid-2000s. Rising field is exactly the regime that produces the high-field effects catalogued in Document 3.

4 The end of Dennard scaling

4.1 The mechanism of failure

The failure is specific and it is a device-physics failure. The chain of reasoning is worth memorising, because every topic in Document 3 hangs off it:

1. Constant-field scaling requires V_{DD} to fall with each node.

2. For the transistor to remain fast, V_T must fall with V_{DD} — otherwise the gate overdrive ($V_{DD} - V_T$) collapses and drive current with it.
3. But in the subthreshold region, drain current falls **exponentially** with gate voltage, and the steepness of that fall is bounded. At room temperature it cannot be better than about 60 mV/decade.
4. So each 60 mV you take off V_T multiplies the off-state leakage current of *every transistor on the chip* by ten.
5. By the early 2000s, with hundreds of millions of transistors per die, that static leakage became comparable to the switching power.
6. Therefore V_T stopped scaling. Therefore V_{DD} stopped scaling. Therefore power density stopped being constant and started rising.

Document 2 derives the 60 mV/decade **limit and shows where it comes from. Document 3 deals with what happens to the device as a consequence. For now, accept it as the binding constraint on the whole industry.**

4.2 What it looked like from outside

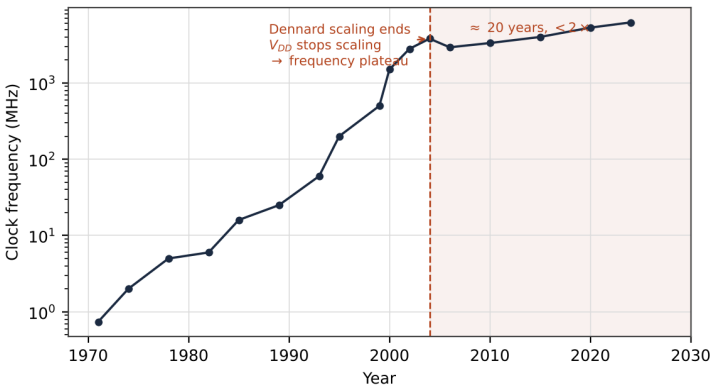


Figure 1.7. *Clock frequency of mainstream desktop processors, 1971–2024. The exponential rise through 2004 is Dennard scaling working. The plateau afterwards is Dennard scaling having stopped: twenty years of continued transistor-count growth bought less than a factor of two in clock rate. Values are representative maximum clocks at introduction; recent values are single-core boost clocks, which flatter the later years — sustained all-core clocks have changed even less.*

With power density rising and no way to dissipate it, frequency scaling stopped dead at around 4 GHz. The industry's response was to spend the transistors Moore's law kept delivering on **parallelism instead of frequency**: multicore processors from around 2005, then GPUs, then domain-specific accelerators. The entire modern computing landscape — multicore CPUs, GPGPU, the AI accelerator boom — is downstream of a threshold-voltage scaling limit.

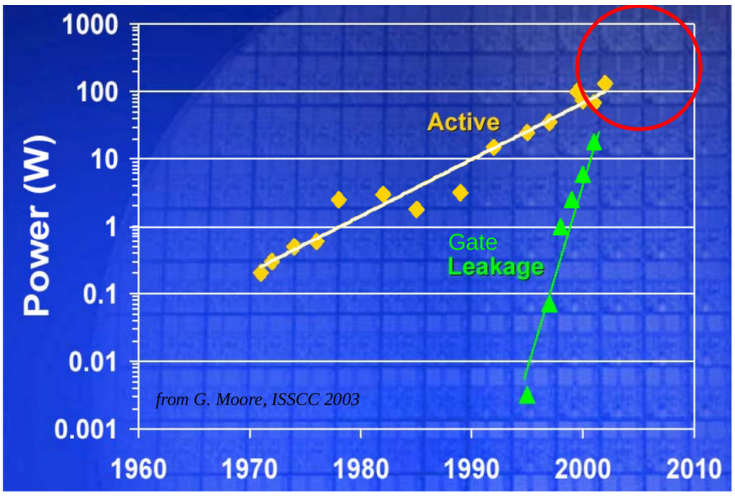


Figure 1.8. *Microprocessor power split into active (switching) power and gate-leakage power, as shown by Gordon Moore in 2003. Leakage was rising far faster than switching power and was on course to overtake it; the high-k/metal-gate change of 2007 (Section 5.2) was the response. Source: G. E. Moore, "No exponential is forever", ISSCC 2003; reproduced from W. A. Doolittle, Georgia Tech lecture slides (2011).*

A related consequence, familiar to anyone with industry experience, is **dark silicon**: if power density is the binding constraint, then beyond some point you can fabricate more transistors than you can afford to switch on simultaneously. Design shifts from "how many transistors can we fit" to "which transistors can we afford to power".

5 What the industry did instead

With the free lunch gone, every subsequent node has required inventing something. This is the sequence, and it is also a preview of the rest of the module.

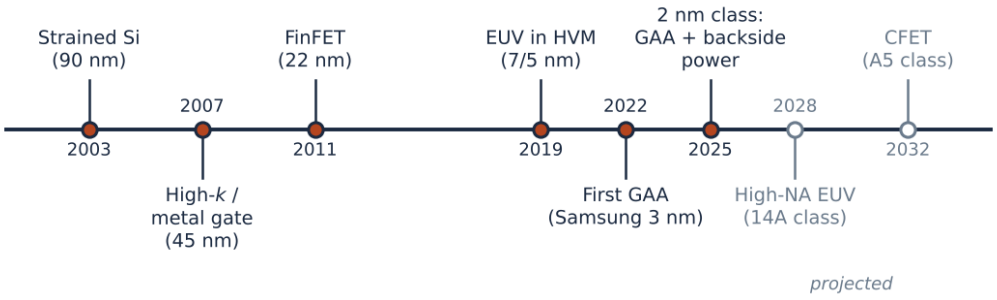


Figure 1.9. *Technology and architecture milestones, 2003–2032. Dates before 2027 are volume-production dates; later entries are industry projections. Samsung was first into GAA production in 2022; 2025 marks the broad transition.*

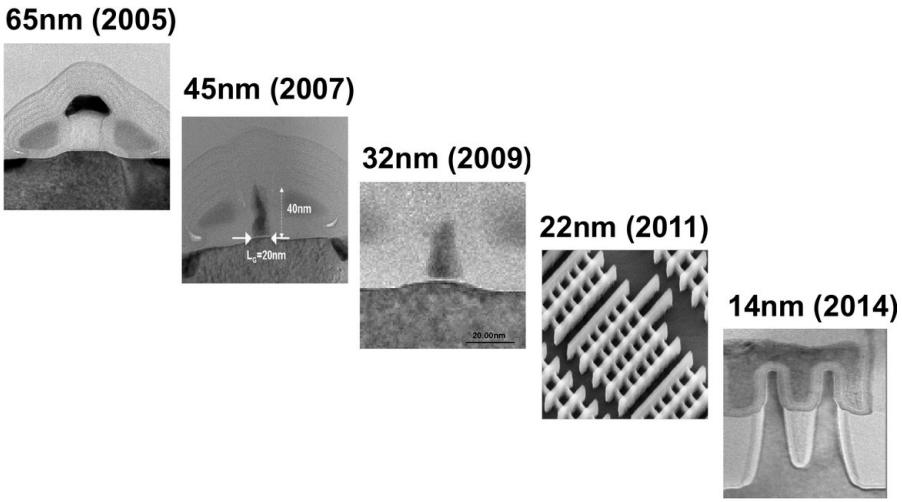


Figure 1.10. *Intel transistor cross-sections, one per generation, from 65 nm (2005) to 14 nm (2014). The planar device gives way to fins at 22 nm. Source: Intel; reproduced from ETH Zürich lecture slides.*

5.1 Strained silicon (90 nm, 2003)

Mechanical strain is applied to the channel, originally via a SiGe layer and later by stressor films and embedded SiGe source/drain regions.

Mechanism — state this correctly. Strain lifts the degeneracy of the conduction-band valleys in silicon and warps the valence band. The useful consequences are a **reduced transport effective mass** and **suppressed intervalley scattering**, both of which raise carrier mobility. Strain does also slightly alter the bandgap, but *that is not the mechanism*, though it is often stated as if it were. Mobility goes up because the band structure changes shape, not because the gap gets smaller.

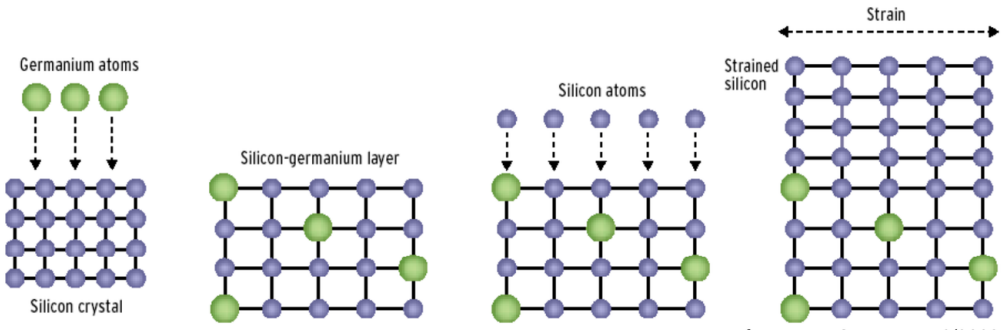


Figure 1.11. *How biaxial strain is introduced: a silicon layer grown on relaxed SiGe takes on the larger SiGe lattice spacing and is stretched. Source: IEEE Spectrum, October 2002; reproduced from W. A. Doolittle, Georgia Tech lecture slides (2011), after O. Brandt.*

5.2 High-k dielectric and metal gate (45 nm, 2007)

By the 90 nm node, SiO_2 gate dielectrics were down to the order of 1 nm — a handful of atomic layers — and electrons were tunnelling directly through them. Gate leakage was becoming a first-order power problem.

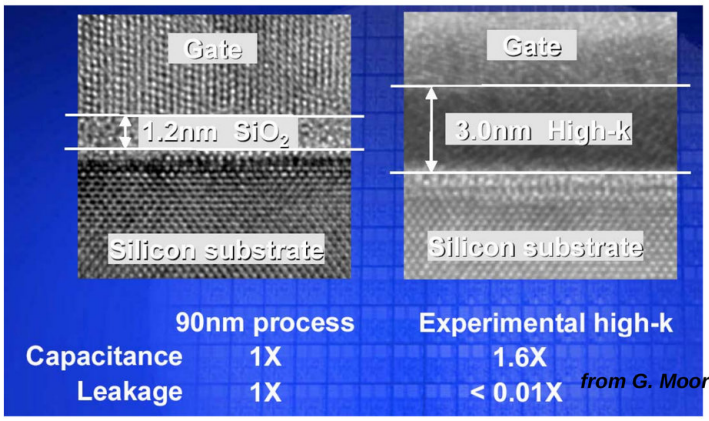


Figure 1.12. *The case for high-k, in two micrographs: a 1.2 nm SiO_2 gate oxide from a 90 nm process beside a 3.0 nm experimental high-k film. The physically thicker film gives 1.6 times the capacitance at less than one-hundredth of the leakage. Source: G. E. Moore, ISSCC 2003; reproduced from W. A. Doolittle, Georgia Tech lecture slides (2011).*

The fix is to keep the *electrical* thickness while increasing the *physical* thickness, which requires a higher permittivity:

$$C_{ox} = \frac{\varepsilon_0 k}{t_{phys}} \qquad \Rightarrow \qquad \mathrm{EOT} = t_{phys}\,\frac{k_{\mathrm{SiO_2}}}{k}$$

Hafnium-based dielectrics (HfO_2, $k \approx 20$–25; HfSiON in practice) give the same capacitance at four to five times the physical thickness, cutting direct tunnelling by orders of magnitude.

A high-k dielectric will not work with a polysilicon gate: Fermi-level pinning at the interface makes the threshold voltage uncontrollable, and remote phonon scattering degrades mobility. So the industry returned to **metal gates**, which it had abandoned in the 1960s. Hence "high-k/metal gate" — the two changes are not independent.

Many candidates were studied — Al_2O_3, ZrO_2, TiO_2 and others. The technology that shipped in 2007, and still ships, is hafnium-based.

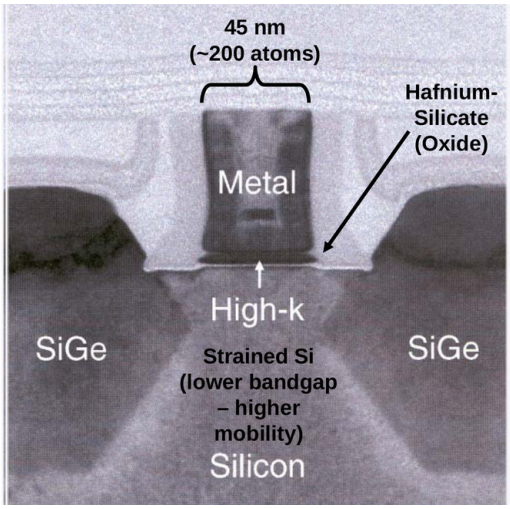


Figure 1.13. *Cross-section of an Intel 45 nm p-channel transistor (2007): metal gate on a hafnium-based high-k dielectric, with embedded SiGe source and drain regions compressing the channel. Note the slide's own label, "lower bandgap – higher mobility": as Section 5.1 explains, the gain comes from the change in band curvature, not from the gap. Image: Intel; reproduced from W. A. Doolittle, Georgia Tech lecture slides (2011).*

5.3 FinFET (22 nm, 2011)

By 22 nm the problem was no longer the dielectric but **electrostatic control**: in a planar device with a channel that short, the drain's field reaches into the channel and competes with the gate's. Document 3 treats this properly.

The structural fix is to wrap the gate around a thin vertical fin of silicon, so the gate controls the channel from three sides instead of one. Intel moved to production FinFETs at 22 nm in 2011 and the rest of the industry followed. FinFET carried logic scaling from 22 nm all the way to the 3 nm generation.

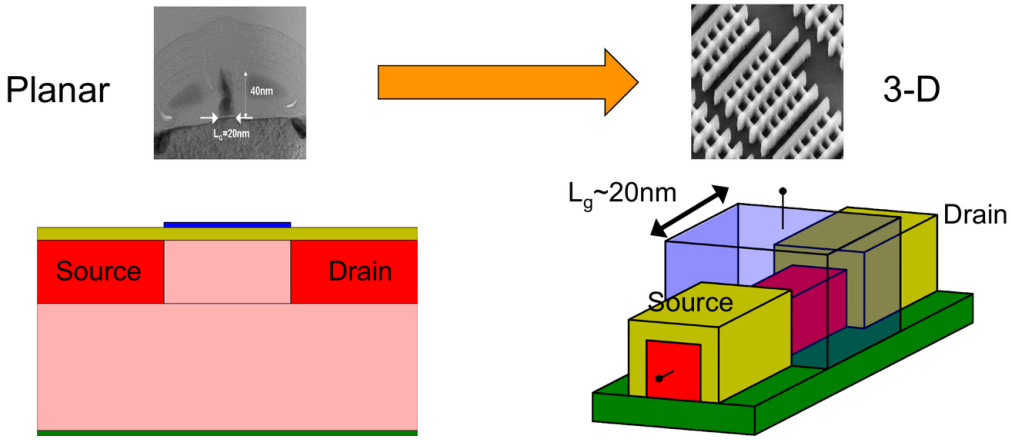


Figure 1.14. *Intel's 2011 change of architecture: a planar transistor (left) and a 22 nm tri-gate, or FinFET, transistor (right), each as a micrograph and as a schematic. Source: Intel; reproduced from ETH Zürich lecture slides.*

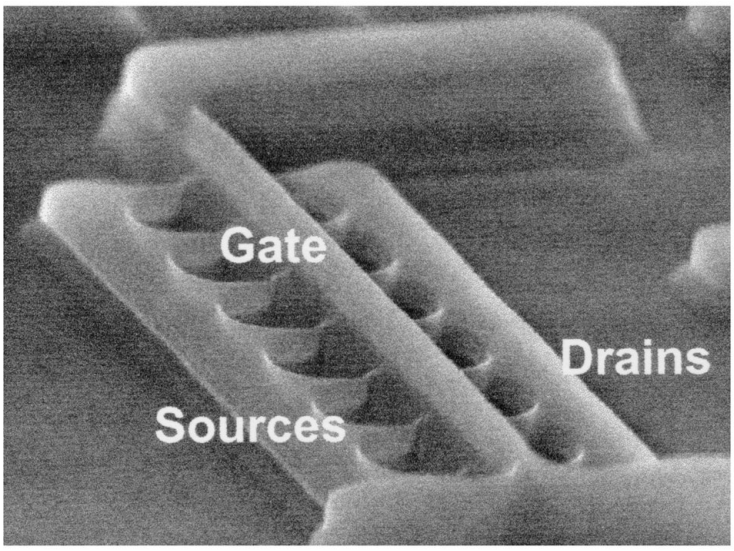


Figure 1.15. *Scanning electron micrograph of an early multi-fin FinFET: a single gate wraps several parallel silicon fins that join common source and drain regions. Source: Intel; reproduced from W. A. Doolittle, Georgia Tech lecture slides (2011), after O. Brandt.*

5.4 Gate-all-around nanosheet (2 nm class, 2025)

The logical conclusion of the FinFET argument is to surround the channel on *all* four sides. In the current implementation the channel is a stack of horizontal silicon sheets with gate material completely enclosing each one. The effective width is set by the sheet width, which is a design choice — unlike FinFET, where width came in quanta of one fin.

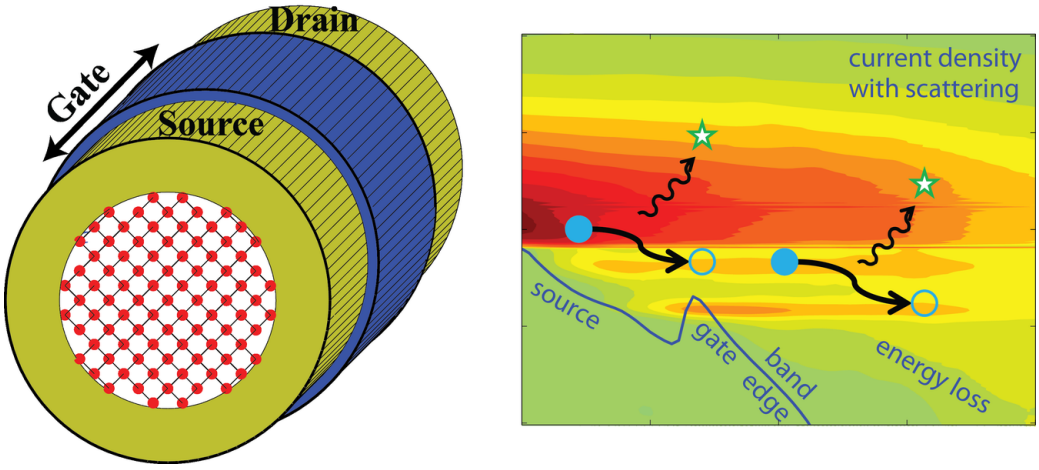


Figure 1.16. *The gate-all-around idea. Left: a nanowire channel completely enclosed by the gate. Right: electrons injected from the source can lose energy by scattering on their way to the drain. Reproduced from ETH Zürich lecture slides.*

Samsung was first into production, with its 3 nm (MBCFET) process in June 2022. The broad transition — and TSMC's and Intel's first GAA nodes — came with the 2 nm class in 2025, the first architecture change at those two companies since the FinFET:

Foundry	Node	Architecture	Volume production
TSMC	N2	GAA nanosheet (first GAA node for TSMC)	Q4 2025, Fab 22 Kaohsiung
Intel	18A	RibbonFET (GAA) **+ PowerVia backside power**	Announced 9 Oct 2025, Fab 52 Arizona
Samsung	SF2	GAA ("MBCFET")	Q4 2025; shipped Q1 2026

TSMC quotes N2 against N3E at 10–15% more performance at equal power, or 25–30% less power at equal performance, with roughly 15% higher density.

5.5 Backside power delivery (2025 onward)

A modern die has of the order of fifteen metal layers above the transistors, carrying both signals and power. Power delivery wants thick, low-resistance wires; signal routing wants fine ones. They compete for the same space, and IR drop in the power network has become a serious limiter.

Backside power delivery moves the power network to the *underside* of the wafer, leaving the front side for signals alone. Intel ships this as **PowerVia** on 18A — the first production node to combine GAA and backside power, which is a deliberately aggressive pairing. **TSMC's N2 does not use it**; TSMC introduces backside power as **Super Power Rail (SPR)** on **A16**, in volume production in H2 2026 alongside **N2P**.

5.6 CFET (research; projected around 2032)

The next structural idea is the **complementary FET**: stack the nMOS and pMOS devices of a CMOS pair vertically on top of each other instead of placing them side by side, which removes the n-to-p spacing from the cell footprint entirely.

Status as of the most recent IEDM reporting: TSMC demonstrated a functional monolithic CFET inverter at a 48 nm gate pitch with backside contacts and interconnect, reporting subthreshold slopes of 74–76 mV/dec for both device types; imec presented a double-row

CFET aimed at an A7 (7 ångström) node and direct backside source/drain contacts at 60 nm pitch; IBM and Samsung showed a stepped-channel stacked FET. imec's roadmap places CFET in mainstream production around the **A5 node, approximately 2032**.

Treat this as research, not product. The gap between a working inverter and a manufacturable node is, historically, about a decade.

6 Lithography: the enabling constraint

No discussion of CMOS technology is complete without the printing step, because for most of the last twenty years lithography has been the gate on everything else.

The wavelength used in production has been stuck at 193 nm (ArF excimer) since the early 2000s. Everything since has been a trick for printing features far below the wavelength:

- **Immersion (2004)** — water between the final lens and the wafer raises the numerical aperture, effectively shortening the wavelength to about 134 nm.
- **Multiple patterning (2010s)** — print a feature, then print again on the offsets; double, then quadruple patterning. Each extra exposure costs a mask, a process cycle, and overlay budget. This is why cost per transistor stopped falling as reliably as it once did.
- **Extreme ultraviolet, EUV (production from 2019)** — 13.5 nm wavelength, generated by vaporising tin droplets with a laser, reflected through an all-mirror system in vacuum. Numerical aperture 0.33.
- **High-NA EUV (now)** — numerical aperture raised to 0.55 with anamorphic optics. Intel has installed the first commercial **ASML Twinscan EXE:5200B** and completed acceptance testing in December 2025, targeting the **Intel 14A** node; ASML points to **2027–28** for high-volume manufacturing use. TSMC A14 and Samsung SF1.4 are the other announced targets.

For this module, the point is that **device architecture and lithography are coupled**. The FinFET was attractive partly because a fin is easier to define with multiple patterning than a precisely doped planar junction. The nanosheet is attractive partly because sheet thickness is set by epitaxy — deposition — rather than by lithography. When you cannot print smaller, you build in the third dimension and let deposition set the critical dimension. That principle recurs throughout this module.

7 The node-name problem

This section exists because the single most common misconception among students entering this module is that "2 nm" is a length.

It is not. It has not been for over a decade. The node name was originally the half-pitch of the smallest metal lines. That correspondence broke down in the 2000s and the names are now pure generation labels, chosen for marketing. The Wikipedia article on the 2 nm process states it plainly: the name "has no relation to any actual physical feature (such as gate length, metal pitch or gate pitch) of the transistors." WikiChip is equally direct: "It does not correspond to any gate length or half pitch."

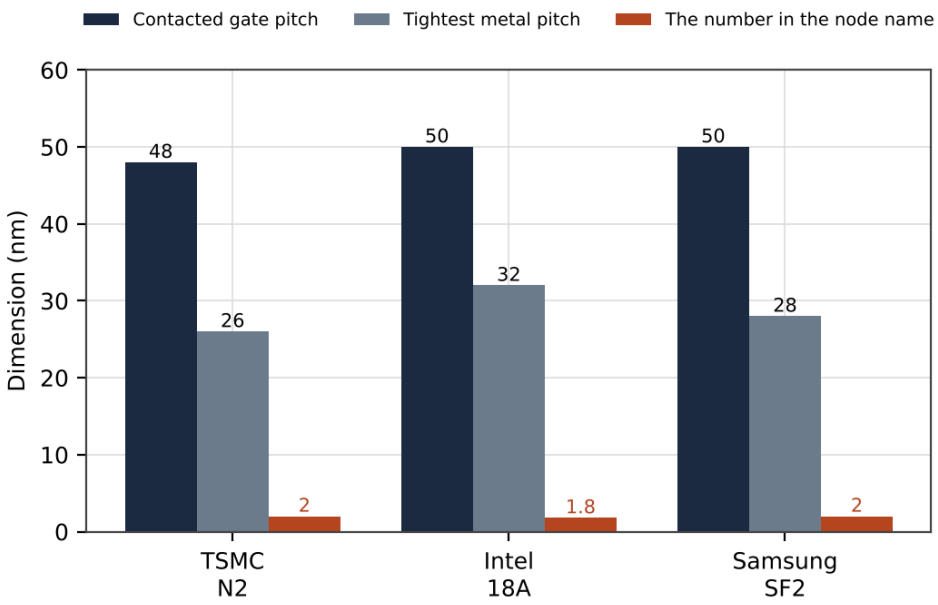


Figure 1.17. *What the node name says, against what is actually on the wafer, for the three 2 nm-class processes now in production. The numbers in the node names are an order of magnitude smaller than any feature they are supposed to describe.*

The real numbers for the three 2 nm-class nodes:

Process	Contacted gate pitch	Tightest metal pitch	Status
TSMC N2	48 nm	26 nm	Volume production Q4 2025
Intel 18A	50 nm	32 nm	Risk Q2 2025, volume Q4 2025, shipped Q1 2026
Samsung SF2	50 nm	28 nm	Volume production Q4 2025, shipped Q1 2026

For the quantity that actually matters to this module — the **physical gate length** — IBM's 2 nm-class gate-all-around prototype used a gate length of about 12 nm with three stacked silicon nanosheets.

So when this module says "nanoscale", the numbers to keep in your head are:

- physical gate length: **of the order of 12–20 nm**
- contacted gate pitch: **about 45–50 nm**
- the number in the node name: **a label**

Hold on to the first of those three. It is the number that makes the physics in Document 2 break.

8 The industry in 2026

Scale. Global semiconductor sales reached **$791.7 billion in 2025**, up 25.6% on the previous year, according to the Semiconductor Industry Association; the World Semiconductor Trade Statistics organisation puts the figure at **$796 billion** on a slightly different basis and projects the market to approach **$1 trillion in 2026** — more than two and a half times the roughly $300 billion of 2010.

Structure. The industry has separated into distinct roles, and this is worth spelling out because it determines where you are likely to work:

- **Pure-play foundries** manufacture to other companies' designs. TSMC dominates the leading edge.
- **Fabless designers** own designs and no fabs — Apple, Qualcomm, NVIDIA, AMD, ARM as an IP licensor.
- **Integrated device manufacturers (IDMs)** still do both; Intel and Samsung both also operate as foundries for external customers.
- **Equipment and materials** — ASML holds an effective monopoly on EUV lithography, which makes it a strategic chokepoint; Applied Materials, Lam, Tokyo Electron and others supply deposition and etch.

What is driving demand. The growth rate above is not a normal semiconductor cycle. It is predominantly AI accelerator and high-bandwidth-memory demand. That matters technically as well as commercially: it is why advanced packaging, memory bandwidth and power delivery — rather than transistor density alone — are now where the hardest engineering sits.

9 Where this leaves us

Pull the threads together, because this is the bridge into the rest of Lecture 1.

Scaling continues, but nothing about it is free. Every node since 2003 has required a materials or structural invention. There is no longer a recipe; there is a sequence of specific engineering responses to specific physical limits.

The limits that bind are device-physics limits. Subthreshold slope. Gate tunnelling. Electrostatic control of a short channel. Carrier transport in a confined, heavily scattered inversion layer. These are not incidental engineering nuisances — they set the roadmap for a trillion-dollar industry.

And classical device physics is now at its own boundary. Consider a single number. The mean free path of an electron in a silicon inversion layer at room temperature is of the order of 10 nm. The physical gate length of a transistor in volume production today is of the order of 12 to 20 nm.

Those two numbers are now comparable. For most of the history of this subject they were not. Around 2000, physical gate lengths in production were of the order of 100 nm, against the same 10 nm mean free path; by 2005 Intel's 65 nm process was already using a 35 nm gate. At 100 nm a carrier crossing the channel scattered many times, which is precisely the assumption that makes mobility, drift velocity and the drift-diffusion equation meaningful quantities.

A carrier crossing a modern channel may scatter once, or not at all. The drift-diffusion framework — the framework in every undergraduate textbook you have read — is being applied at the edge of, and sometimes beyond, its domain of validity. Transport is *quasi-ballistic*.

That is the subject of Document 2: what the classical picture says, stated compactly and in one notation, and exactly where and why it stops being true at these dimensions.

Sources and further reading

Current industry status (all accessed October 2026):

- TSMC N2 volume production, architecture and figures: Tom's Hardware, *TSMC begins quietly volume production of 2nm-class chips*. https://www.tomshardware.com/tech-industry/semiconductors/tsmc-begins-quietly-volume-production-of-2nm-class-chips-first-gaa-transistor-for-tsmc-claims-up-to-15-percent-improvement-at-iso-power
- Intel 18A production and comparison with N2: Tom's Hardware, *Intel's 18A production starts before TSMC's competing N2 tech*. https://www.tomshardware.com/pc-components/cpus/intels-18a-production-starts-before-tsmcs-competing-n2-tech-heres-how-the-two-process-nodes-compare
- Node dimensions and production dates: Wikipedia, *2 nm process*. https://en.wikipedia.org/wiki/2_nm_process
- Node naming: WikiChip, *Technology node*. https://en.wikichip.org/wiki/technology_node
- CFET results and roadmap: eeNews Europe, *IEDM: CFETs make progress at 5nm and 7angstrom*. https://www.eenewseurope.com/en/iedm-cfets-make-progress-at-5nm-and-7angstrom/; imec, *Scaling monolithic CFET across multiple logic technology nodes*. https://www.imec-int.com/en/articles/performance-boosters-scale-monolithic-cfet-across-multiple-logic-technology-nodes
- High-NA EUV: Bits&Chips, *Intel completes acceptance testing of ASML's EXE:5200B high-NA EUV scanner*. https://bits-chips.com/article/intel-begins-acceptance-testing-of-asmls-exe5200b-high-na-euv-scanner/
- Market size: Semiconductor Industry Association, *Global annual semiconductor sales increase 25.6% to $791.7 billion in 2025*. https://www.semiconductors.org/global-annual-semiconductor-sales-increase-25-6-to-791-7-billion-in-2025/; WSTS. https://www.wsts.org/76/103/Global-Semiconductor-Market-grows-26-in-2025-to-796B
- Transistor counts: Wikipedia, *Transistor count*. https://en.wikipedia.org/wiki/Transistor_count

History:

- Wikipedia, *Shockley Semiconductor Laboratory*. https://en.wikipedia.org/wiki/Shockley_Semiconductor_Laboratory
- G. E. Moore, "Cramming more components onto integrated circuits", *Electronics*, vol. 38, no. 8, pp. 114–117, 19 April 1965.

Primary literature worth reading:

- R. H. Dennard *et al.*, "Design of ion-implanted MOSFET's with very small physical dimensions", *IEEE J. Solid-State Circuits*, vol. 9, no. 5, pp. 256–268, 1974. *The scaling paper. Still worth reading in full; it is short.*
- M. Bohr, "A 30 year retrospective on Dennard's MOSFET scaling paper", *IEEE SSCS Newsletter*, vol. 12, no. 1, pp. 11–13, 2007.

Figure credits. Figures 1.5, 1.7, 1.9 and 1.17 were generated for this document from the data sources cited above. Figure 1.4 is Moore's 1965 figure (image courtesy of Intel). The

other figures are reproduced from earlier course material for this module — lecture slides by W. A. Doolittle (Georgia Tech, 2011), ETH Zürich lecture slides, and S. Levkov's NJIT ECE 271 slides — with the original source, where known, given in each caption. Copyright in reproduced figures remains with the original owners.